

YELABUGA TATARSTAN, Russian Federation

WMO: 285060

Lat: 55.759N

Lon: 52.044E

Elev: 91

StdP: 100.24

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.4	-25.1	-31.6	0.2	-28.2	-28.1	0.3	-24.9	13.1	-6.9	10.5	-6.2	2.1	230	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	31.3	20.4	29.2	20.0	27.4	18.8	21.7	28.8	20.7	27.5	19.8	25.9	2.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.3	24.6	18.4	13.4	23.7	17.5	12.6	23.1	63.6	28.9	60.2	27.8	57.1	26.0	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.8	7.6	DB	-31.3	33.1	4.4	2.3	-34.5	34.8	-37.0	36.1	-39.5	37.5	-42.7	39.2	(4)
(5)			WB	-31.5	23.6	4.3	2.6	-34.6	25.5	-37.1	27.0	-39.6	28.5	-42.7	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-11.4	-10.8	-4.0	5.2	13.1	17.8	20.1	17.7	12.0	4.8	-2.9	-8.8	(6)
(7)		DBStd	12.52	7.17	7.16	4.86	5.59	5.15	4.41	3.90	4.38	4.58	4.66	5.91	7.21	(7)
(8)		HDD10.0	3047	663	583	435	164	29	3	0	1	29	171	386	583	(8)
(9)		HDD18.3	5246	921	816	693	394	174	62	27	66	197	418	636	842	(9)
(10)		CDD10.0	1036	0	0	0	20	126	237	314	239	89	11	0	0	(10)
(11)		CDD18.3	194	0	0	0	0	12	47	83	46	7	0	0	0	(11)
(12)		CDH23.3	1810	0	0	0	3	136	415	777	430	49	0	0	0	(12)
(13)		CDH26.7	563	0	0	0	0	19	121	260	155	8	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.8	2.8	3.0	3.1	3.0	2.6	2.3	2.5	2.6	3.1	3.0	2.8	(14)
(15)	Precipitation	PrecAvg	544	38	29	27	41	45	52	70	56	58	55	42	39	(15)
(16)		PrecMax	773	80	77	88	139	109	109	169	151	160	111	103	76	(16)
(17)		PrecMin	362	6	0	1	10	5	4	23	8	6	1	11	2	(17)
(18)		PrecStd	114	20	17	18	28	26	26	38	34	42	27	22	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	2.4	9.0	23.6	28.7	32.6	34.1	34.5	27.8	19.3	10.7	3.2	(19)
(20)			MCWB	1.2	0.9	5.2	13.2	17.6	20.2	20.7	21.5	16.7	12.3	9.4	2.0	(20)
(21)		2%	DB	1.0	1.5	5.5	20.0	26.8	29.8	31.5	30.7	24.4	16.4	7.8	2.0	(21)
(22)			MCWB	0.2	0.4	2.5	11.4	16.6	19.9	20.5	20.3	15.2	10.9	6.3	1.1	(22)
(23)		5%	DB	-0.3	0.6	3.6	17.3	24.7	27.7	29.6	27.8	21.9	13.7	5.5	1.1	(23)
(24)			MCWB	-1.0	-0.2	1.5	10.1	15.5	18.9	20.5	19.3	14.8	9.8	4.4	0.3	(24)
(25)		10%	DB	-1.8	-1.0	2.2	14.3	22.3	25.6	27.6	25.2	19.2	11.5	3.6	0.1	(25)
(26)			MCWB	-2.5	-1.8	0.7	8.7	14.3	17.9	19.8	18.2	13.7	8.8	2.6	-0.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.2	5.3	14.0	19.0	21.8	24.1	22.8	18.7	13.3	8.7	2.4	(27)
(28)			MCDWB	2.1	1.9	8.6	22.5	26.2	29.6	28.9	31.2	25.1	16.8	10.2	2.8	(28)
(29)		2%	WB	0.3	0.6	2.9	12.3	17.4	20.6	22.1	21.3	16.5	11.7	6.6	1.2	(29)
(30)			MCDWB	0.9	1.3	4.9	18.7	24.8	27.7	28.7	28.9	21.7	15.3	7.6	1.8	(30)
(31)		5%	WB	-0.9	-0.2	1.8	10.5	16.2	19.6	21.1	20.0	15.3	10.2	4.5	0.3	(31)
(32)			MCDWB	-0.3	0.7	3.2	16.5	23.6	26.3	27.7	26.4	20.1	13.0	5.4	0.9	(32)
(33)		10%	WB	-2.5	-1.9	0.9	8.9	15.0	18.6	20.3	18.8	14.1	9.0	2.7	-0.5	(33)
(34)			MCDWB	-1.8	-1.0	2.1	14.3	21.2	24.6	26.3	23.9	18.7	11.4	3.4	0.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	6.7	7.2	8.8	10.2	9.6	9.8	9.4	8.3	6.0	4.3	5.1	(35)
(36)			MCDBR	4.8	4.7	7.3	13.0	13.6	12.6	12.9	13.2	12.1	10.6	4.7	3.4	(36)
(37)			MCWBR	4.6	4.4	5.4	6.6	6.3	5.5	5.0	5.5	5.7	6.3	4.2	3.3	(37)
(38)		5% WB	MCDWR	5.1	4.5	6.4	12.4	13.1	11.7	11.2	12.2	10.7	9.3	4.6	3.4	(38)
(39)			MCWBR	5.0	4.3	5.0	6.8	6.5	5.7	5.2	5.9	6.0	6.1	4.3	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.321	0.373	0.414	0.398	0.392	0.412	0.436	0.381	0.356	0.319	0.294	(40)	
(41)		taud	2.030	2.025	1.957	2.087	2.283	2.350	2.298	2.205	2.335	2.284	2.164	2.016	(41)	
(42)		Ebn at Noon	568	712	762	798	842	853	824	771	769	682	550	464	(42)	
(43)		Edh at Noon	67	99	136	139	122	115	119	123	94	77	59	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.71	1.71	3.10	4.40	5.44	5.90	5.79	4.50	2.79	1.27	0.53	0.39	(44)	
(45)		RadStd	0.10	0.20	0.33	0.39	0.49	0.54	0.52	0.40	0.34	0.25	0.10	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	N/A

Nomenclature: See separate page